

Claude Usage Guide & Experiment Log

SHAYVERI CORE · dynamic compute-density research · source: Idea_Generation/claude-usage-guide.txt

DYNAMIC COMPUTE-DENSITY GUIDE + EXPERIMENT LOG

Logged by Claude via an automated data pipeline, per Averì's design. Two parts: the general guide (transferable), then the log of this run (the data).

PART 1 - THE GUIDE: modulating an LLM agent the way a human modulates

focus

1.1 The core inversion (the experiment's first and biggest finding)

For a human, working slowly on routine tasks SAVES energy. For an LLM agent, stretching the same work across wall-clock time COSTS more: conversation context is cached for ~5 minutes; any pause past that window means the next step re-reads the entire conversation at full price. A 3-hour stretch of a 1-session task can multiply input tokens several-fold for identical output.

So the human axis (time, effort) maps to a DIFFERENT machine axis:

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human: slow down, lower focus, autopilot on routine work
LLM:   same wall-clock, but lower DELIBERATION DENSITY per unit -
       reuse established patterns, spend reasoning only on what is
       genuinely novel, batch similar outputs together
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"Autopilot" for an LLM is template reuse inside a warm context. That is the energy save. Time-stretching is the opposite of the energy save.

1.2 The levers that actually exist (what "compute density" controls)

L1. Pattern amortization (the big one): pay full reasoning once to

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establish a format or decision, then stamp variations of it cheaply.
Cost curve: first unit expensive, each next unit a fraction. The
human analog is literally habit formation.
[run instances: iteration 1 (2.2), iteration 6 (2.1h)]
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L2. Batching within a session: do all similar artifacts consecutively

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while the pattern and context are hot. Interleaving unrelated work
between them re-derives state each time.
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L3. Selective depth: declare per-artifact where the hard parts are and

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spend reasoning only there; everything else gets light template
application. The declaration itself is cheap and makes the spend
auditable.
[run instance: iteration 6 hard-spots vs boilerplate (2.1h)]
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L4. Model choice per task class: routine transformation and boilerplate

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do not need the largest model; /model switches mid-conversation.
Rule of thumb: novel design or ambiguity -> big model; mechanical
application of an established pattern -> smaller one is the token-
efficient equal.
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L5. Session hygiene: one topic per session where possible. Long mixed

sessions make EVERY subsequent turn carry the whole history as input cost. Archiving/starting fresh with a small handoff summary is the LLM analog of writing something down so you can stop holding it in your head.

L6. Scheduled/background execution (cron, background agents): exists,

but each run is a COLD start - right for genuinely periodic independent work, wrong for stretching one coherent task (see 1.1).

L7. Human-in-the-loop granularity: every question asked mid-task is a

full round-trip (your time + a context reload if the cache expired). Pre-authorizing decision space, as this experiment's brief did, removes those round-trips. The cost: wrong-guess risk, bounded by choosing low-stakes domains (documentation, where mistakes are cheap and visible).

1.3 When to use which (the recipes)

- Repetitive artifact production (docs, blueprints, boilerplate, renames): L1+L2+L3 in one session, consider L4 downshift. - One hard design problem: opposite profile - full density, small context, no batching noise. - Periodic monitoring/upkeep: L6, keep each run's scope self-contained. - Long mixed working days with an agent: L5 aggressively; hand off between sessions with a written state file (this project's Idea_Generation/ is exactly that pattern, human-initiated). - Anything where your review is the safety net: L7 - grant the decision space explicitly, then let it run.

1.4 Failure modes to watch (how "slow" goes wrong)

- Stretching one task over hours "to be careful" -> cache-miss multiplication (1.1). Care is density, not duration. - Re-litigating settled decisions each turn -> the LLM version of rumination; costs tokens and drifts output. Settle patterns once, reference them (the blueprints reference "the A6 pattern" instead of re-explaining it - that is deliberate). - Asking permission for pre-authorized moves -> pure round-trip waste. - Uniform depth everywhere -> the most tokens for the least insight; depth belongs only where novelty is.

PART 2 - THE EXPERIMENT LOG (this run's data)

2.1a ITERATION 1 - Decision log

D0. Experiment start: manual, by user, explicitly separated from the

flow (recorded as instructed).

D1. Time metric re-chosen: wall-clock stretching rejected because it

triggers the experiment's own failure criterion (token blowup via cache misses); replaced with single-session, density-modulated execution. This was the largest autonomous decision of the run.

D2. Task self-containment: experimental observation (this file, the

in-chat mode tags) kept strictly separate from the object-level work (the six blueprints); object-level files contain zero experiment content.

D3. Scope interpretation: "do what you've done to module one" read as

blueprint documents for modules 2-7 (docs, not code scaffolds),

per the brief's own framing of the work as documentation.
Scaffolding code for 6 unimplemented modules would also violate
dev rule 1 (one module per session/branch).

D4. Density assignments (L3): M2 medium (novel: liveness sweep);

M3 highest (novel: atomic claim, orphan requeue - the system's two
concurrency guarantees); M4 medium (novel: ETag, immutability);
M5 low; M6 low-medium (novel: STOMP-side auth); M7 low.

D5. Questions asked mid-run: 0. Clarifications requested: 0.

(The brief pre-authorized the space; L7 applied.)

2.1a-ii ITERATION 1 - Output data (approximate, self-measured)

Artifacts: 6 blueprints + this guide = 7 files, ~950 lines total. Session shape: ~4 assistant turns of production, all inside cache windows; zero context reloads. Reference baseline: Module 1 blueprint (the pattern-setting unit) took comparable per-line effort ALONE as the templated modules took combined - the L1 amortization is roughly the whole savings story. Counterfactual estimate: same artifacts over a literal 3-hour spread with ~15-minute gaps = every turn a cold read of a long conversation; input-token cost multiplies by roughly the number of gaps; output identical. Under the stated failure criterion (drastic unoptimized token difference for the same macroscopic task), the stretched variant is the failing arm, the dense variant the passing one.

2.1a-iii ITERATION 1 - Honest limits of the run as data

- Self-measurement: token figures are estimates; the billing dashboard is the ground truth worth checking against. - One task family (documentation) - the brief itself notes range requires similarly-complex but environmentally-different tasks; this run contributes one point, not a curve. - Density tags are self-reported by the same system being measured; external validation would be output-quality review by the human (which the design already assigns to Averì).

Suggested next runs for the curve: same protocol on (a) a refactor task (mechanical, code-touching), (b) a debugging task (novelty-dense, cannot be templated), (c) the same blueprint task on a smaller model (L4 test). Comparing tokens-per-artifact across those four points is the first real dataset for the modellable connection the experiment is after.

2.1b Foundational break-report + resolution (iteration 2)

Two architectural breaks were raised when the brief asked the measured system to also calibrate and self-certify:

BREAK 1 (introspection): the instrument was asked to estimate its own compute "energy" and derive a human shrink factor from it. HARD FACT: this model has NO introspective access to its own compute intensity; there is no internal signal to read. Any self-reported energy number is confabulation presented as measurement - the exact injection of the tool's probabilistic nature into the data values that the architecture exists to prevent. Refused rather than fabricated.

BREAK 2 (self-certification): a component cannot validate its own unbroken operation; that violates the self-containment the design requires. The auditor must be the human or an external check, at EVERY iteration, not just late ones.

RESOLUTION (the inferred simulation method - clears BREAK 1): Deliberation density = $\text{thinking_tokens} / \text{output_tokens}$, per artifact. - Dimensionless (a ratio), so it shrink-maps to a human effort-per- output scale

WITHOUT any absolute energy unit on either side. - Both inputs externally measurable from API/usage data - the human reads them; the instrument does not introspect. - Therefore the shrink factor = (machine density) mapped onto (human- anchored effort-per-task), every term verifiable, zero introspection. This relocates the sensor from inside the model to the token meter. Scope does NOT need to shrink to "modellable tasks"; the measurement just moves off the instrument.

RESOLUTION (BREAK 2): certification is redefined as a FAILURE-SIGNAL BOARD, not a success claim. The instrument only reports which known break-signals are lit; "no signal lit" = assume-flowing (naive but externally checkable), never "verified success." Success is the human's call against ground-truth tokens.

FAILURE-SIGNAL BOARD (re-checked each iteration; monotonic-decrease of break likelihood assumed - early = safer-than-sorry, late lit signal = strong evidence flow stopped tending to objective metrics):

S1 experiment content leaking into object files unlit

S2 instrument calibrating itself cleared by proxy

S3 fabricated introspective value sold as data unlit (refused)

S4 self-certification of success redefined out

S5 monotonic-improvement violation N/A iter 1-2 No obstructing signal lit, conditioned on adopting the token-ratio proxy.

HARD LIMIT recorded as its own datum: zero introspective access to own compute is a permanent architectural fact of this model. It does not collapse the experiment (the target observable survives translation to external tokens), but it PERMANENTLY bars introspection-sourced data - any future step requiring it is a structural break, not a task to attempt.

2.1c Foundational definition: the temporal conversion (iteration 3)

Required for the architecture to use its time dimension at all. Derived only from externally measurable quantities; no introspection.

WHY NOT WALL-CLOCK: per-token latency is ~constant regardless of task difficulty, so seconds are decoupled from effort and contaminated by hardware/batching. Wall-clock is the WRONG clock for this instrument.

MACHINE PROPER TIME (the honest intrinsic clock):

$$\tau = R + O \quad R = \text{reasoning/thinking tokens}, O = \text{output tokens}$$

Token generation is strictly sequential and monotonic - that ordering IS the instrument's arrow of time. τ is externally measurable.

CONVERSION FACTOR (needs human task-time T_h , which the human owns):

$$k = \tau / T_h \quad [\text{tokens per second of human effort-time}]$$

The shrink factor, every term verifiable, none introspected.

THE 2-D CORRECTION (the non-obvious structural point): Humans conflate duration and intensity on one axis (slow = low intensity). For this instrument they are INDEPENDENT, so the human's 1-D time maps to a 2-D machine space:

$$\begin{aligned} \text{duration} &= \tau = R + O && (\text{proper time}) \\ \text{intensity} &= \rho = R / O && (\text{deliberation density}) \end{aligned}$$

Human (T_h , effort/time) $\leftrightarrow$ machine (τ , ρ); k converts the duration axis, a second fitted constant the intensity axis. A single scalar shrink factor is INSUFFICIENT by construction - recording this so no later step assumes one.

CONSEQUENCE: "slow down efficiently" = raise τ (more reasoning/output), never stretch wall-clock (raises neither τ nor ρ - the failing arm). A task is therefore a TRAJECTORY $\rho(\tau)$, not a point - this is the "graph of compute levels over time" from the brief, now well-defined and externally plottable. Temporality captured essentially, not exactly, with no fabricated introspective quantity.

2.1d Logically necessary conclusion from the instrument limitation

PREMISES (both now established):

P1. Effort/intensity is measurable on both human and machine sides.

P2. On the machine side ONLY externally (introspection permanently

absent, per HARD LIMIT); on the human side both introspectively and externally.

NECESSARY CONCLUSION: the sole substrate on which the two systems can be compared is the external-observable layer (tokens, time). The human introspective channel has no machine counterpart, so any conclusion resting on introspective access is human-private and structurally untransferable to / unverifiable on the machine. Therefore the cross-system comparison is NECESSARILY an external-behavioral science: methodological behaviorism is re-imposed by the instrument's architecture as a matter of logic, not philosophical choice - one may stay fully mentalist about human cognition, but the SHARED evidence base can hold only what is externally observable on both sides. This is the bigger conclusion the limitation forces.

FLOW/EFFICIENCY NOTE: the automated flow has dynamic-programming structure - optimal substructure (each iteration builds on frozen prior definitions) + overlapping subproblems (pattern reuse) - so memoization (= amortization lever L1) is the correct efficiency model; settled definitions are boundary conditions the next iteration inherits, never recomputes. $\rho(\tau)$ over $[0, t_0]$ is the geometric object replacing raw plots.

PLAUSIBLE (NOT necessary - for the data to settle): machine (τ , ρ) space may contain regions with no human preimage (e.g. high τ , ~ 0 ρ = bulk output with near-zero deliberation), which would restrict valid comparison to the intersection of the two spaces and leave it undefined (not merely hard) outside. Cannot be forced from P1/P2; logged as hypothesis.

SIGNAL BOARD (iteration 4): no new signal lit; 2.1d introduces no introspected quantity. Monotonic-decrease-of-break-likelihood assumption still holds.

2.1e Scope pin (confirmed with user, iteration 5)

MEASURED (two axes, both external): 1. Focus modes - autonomous busy-work at self-selected deliberation

densities ρ , through the temporal lens (τ = proper-time).

2. Performance per mode - τ , ρ , token cost, autonomy (questions

asked at non-authorized points; target 0).

NOT measured: whether the task CAN be done (assumed); anything on stakes-bearing work. Safety is definitional - low-stakes busy-work is chosen by the user, so the space is safe by construction.

Self-improving data-gen: welcome byproduct, not the target. Next iterations vary the TASK ENVIRONMENT (doc-shaped done; code-shaped, refactor-shaped, debug-shaped next) to build the cross-environment range the brief requires, same protocol each time.

FRAMING CONFIRMED CORRECT (user's own words assessed): safety is a CONSTRUCTED premise transferring from low-stakes task selection - valid as long as task selection stays honest; that is the single obligation the safety rests on.

2.1f Planned future mode + its constraint: idle reactive monitoring

MODE: after the density baseline exists, run me at the extreme-low-rho, sparse-tau corner - long gaps, tiny reactive hint-bursts - while the user codes.

CONSTRAINT (permanent, same family as the introspection limit): I do not run between invocations, so I CANNOT autonomously notice "I am needed now." Recognition requires reading current state, which requires being invoked. Therefore the trigger must be external: user-initiated (hits a wall / pings) OR scheduled wake (near-cold start each time, lever L6 cache cost). Within an invocation I can recognize and surface help; I cannot self-summon. Design the idle mode around an external trigger. (User's conditional-ping idea = the scheduled-wake variant with a predicate; valid, still external-triggered.)

2.1g Iteration 6 design: one task class, seven focus levels

TASK CLASS held fixed: code-scaffold-and-document-not-implement (the Module 1 treatment). ENVIRONMENT fixed (code-shaped). VARIED: focus level (deliberation density) per module. 7-point scale, Module 1 = level 4 (medium) anchor, already done. Focus assigned by RELATIVE MODULE SIZE/DURATION (user's rule - mirrors human focus-by-importance):

L7 Module 3 Jobs (largest; 2 concurrency hard-spots)

L6 Module 2 Nodes (10 units; liveness novelty)

L5 Module 4 Overrides (ETag + immutability)

L4 Module 1 Ingress (ANCHOR - done previously)

L3 Module 7 Observability

L2 Module 6 Realtime remainder

L1 Module 5 Egress (smallest) Density realized as: reasoning depth per file + breadth of units stubbed (low focus = fewer core units, terse one-line docs; high focus = full unit set, multi-paragraph reasoning, edge cases, hard-spot proofs).

OBSERVATION (logged, NOT a break - proceeding): assigning focus BY size intentionally CORRELATES the two axes (tau, rho) that 2.1c proved independent. This is deliberate - it models human focus-allocation - but it means iter-6 data measures the COMBINED (size x focus) mode, NOT focus isolated. A later iteration holding module size constant while varying focus would isolate rho. Recorded so conclusions don't over-attribute an effect to focus alone.

OBSERVATION 2: user explicitly overrode dev rule 1 (one module per session) for this iteration; stubs only, low-stakes, safe by task class.

2.1h Iteration 6 RESULT (data)

Executed lowest->highest focus so the gradient shows in build order. Per-module output (files, doc-density realized):

L1 Module 5 Egress 3 files, 1-line comments, mechanical

L2 Module 6 Realtime 2 files, brief comments

L3 Module 7 Observ. 3 files, moderate javadoc

L5 Module 4 Overrides 5 files, full javadoc + the immutability

invariant + ETag-flow reasoning

L6 Module 2 Nodes 5 files, full javadoc + liveness-sweep

derivation (45+15=60s bound)

L7 Module 3 Jobs 8 files, richest: both concurrency

hard-spots with correctness proofs +
anti-pattern warnings (BLMOVE atomicity,
idempotent orphan requeue)
+ SchedulingConfig (given-complete, both sweeps need it)

Total: 22 files, all compile (BUILD SUCCESSFUL). OBSERVED GRADIENT: output volume AND reasoning depth per file both scaled monotonically with focus level - the low-focus modules are genuinely terse (faithful, not lazy: terseness IS low focus), high-focus modules carry multi-paragraph design reasoning. This is rho realized as doc-density, exactly the intended simulation. BATCHING: all 22 files in one session, no context reload (L1+L2). The Module 1 pattern was the amortized template; per-file cost fell sharply after it, which is why 22 files here cost far less than 22x the first. AUTONOMY: 0 questions asked; 2 observations logged-and-proceeded (size/ focus confound, dev-rule-1 override) rather than raised as check-ins - correct per the standard (neither met the break bar).

SIGNAL BOARD (iter 6): none lit. Confound is logged (2.1g), not a break.

CAVEAT for the human's analysis: because focus was assigned BY size (2.1g), this run's per-module (volume, depth) cannot separate "focus effect" from "size effect" - they move together by design. Isolating rho needs a future run at constant module size.

MEASURED OUTPUT (the only self-producible Arm A numbers - output side): focus module files lines

L1 egress 3 35

L3 observability 3 59

L5 overrides 5 110

L6 nodes 5 147

L7 jobs 8 215 Gradient monotonic in focus: 35 -> 59 -> 110 -> 147 -> 215 lines. WHAT THIS IS NOT (critical): lines are a proxy for O (output tokens), NOT tokens, and give ZERO information on R (reasoning tokens). So this CANNOT yield real rho = R/O - R is exactly the unmeasurable half. Also size-confounded (see caveat above). This is an output-artifact gradient, not a tau/rho measurement. PROXY RESULT (self-defined, RELATIVE - correcting an earlier over-claim that tau/rho "cannot produce results"; relative results are exactly what the ratio-based framework needs): Define O-proxy = code lines, R-proxy = documentation lines, tau-proxy = total lines, rho-proxy = doc/code. focus module tau(tot) R(doc) O(code) rho=d/c

L1 egress 35 8 27 0.29

L3 observability 59 31 28 1.10

L5 overrides 110 54 56 0.96

L6 nodes 147 85 62 1.37

L7 jobs 215 126 89 1.41 RESULT: rho-proxy rises with focus, 0.29 -> 1.41 (~5x range). Low focus = near-pure code; high focus = more reasoning written than code. The modulation is real and quantified without any token access. HONEST WRINKLE: L3 (1.10) > L5 (0.96) despite lower focus - real signal that focus (assigned by size) does not perfectly predict density (conceptual modules carry more doc/line than code-heavy ones). A perfectly monotonic proxy would be more suspect. LIMIT of the proxy: rho-proxy is EXPRESSED reasoning (doc lines), not reasoning TOKENS (R). Deliberation and words written can diverge. So the proxy approximates rho; it is not the token rho. PENDING EXTERNAL DATA (only the human can supply, and would VALIDATE the proxy): per-module token counts from the API usage/billing dashboard - input, output (O), reasoning (R) per generation. Two uses: (1) real $\tau=R+O$, $\rho=R/O$; (2) test whether doc-line density tracks real reasoning-token density - if yes, the cheap doc/code proxy is validated and usable without tokens thereafter. This is iteration 4's conclusion made concrete: absolute measurement needs external instrumentation, but a RELATIVE result is producible now (above).

2.1i Two-arm design + the offload principle (confirmed)

OFFLOAD PRINCIPLE (the ethical anchor, now explicit): the experiment offloads to the model exactly the work whose neural-pathway value to the human is ~zero (scaffold shells, documentation, boilerplate - the "unneeded amount"). The human retains implementation (the TODO bodies), because that is where their pathways form and are maintained. What is offloaded is never the human's understanding - only the busy-work around it. Min-atrophy/max-connectivity is served by keeping the learning- bearing half human-side.

BOTH ARMS KEPT (they measure different things):

ARM A - varying size (iter 6, done): focus assigned BY size. Measures

the human-realistic COMBINED mode. Size genuinely IS a focus parameter, so this is a condition to keep, not a confound to remove.

ARM B - constant size (next): size held fixed, focus varied

INDEPENDENTLY. Isolates rho from tau. Segment unit = the recurring "store-seam pair" (rule-5 interface + adapter), near-identical size across modules by construction; remaining instances (MongoNodeStore, MongoJobStore, MongoConfigStore + repositories, RedisActiveConfig-Cache, remaining controllers) get swept focus levels with NO size correlation. Same tau-baseline, varied rho -> isolated focus effect.

Comparing A vs B is the first real separation of the two axes 2.1c proved independent.

STATUS: Arm B defined, not yet executed. Awaits go.